

Satyam Mittal

+91-9042017967 | mittal.satyam646@gmail.com | GitHub | LinkedIn

EDUCATION

- **Bachelor of Technology in Computer Science and Engineering** 2023-2027
Vellore Institute Of Technology, Vellore CGPA: 9.16
- **High Secondary (XII Standard)** May-2023
DAV Sushil Kedia Vishwa Bharati Percentage: 88.4

PROJECTS

- **Vision-Based Depth & Horizon Estimation for Rocket Payload** | [LINK](#) January 2026- Ongoing
Monocular Depth Mapping and Horizon-Based Attitude Analysis
 - Vision-based system to estimate scene depth and analyze horizon orientation for a rocket payload.
 - Implemented a monocular depth model using Python and PyTorch, and built a horizon detection pipeline using OpenCV-based image segmentation for roll/pitch estimation.
 - Tested on Raspberry Pi, generating stable depth maps and estimating relative altitude during experimental trials.
- **Early Disease Detector using Human Nail** | [LINK](#) May - June 2025
Early Stage Disease Diagnosis using Nail Image Processing
 - Developed a deep learning-based system to detect early-stage diseases from human nail images.
 - Fine-tuned a pre-trained Xception model using TensorFlow/Keras with data augmentation to improve generalization.
 - Achieved 87% training accuracy and 89% validation accuracy on the curated dataset.
- **Vidzy** | [LINK](#) April - June 2025
A modern platform to upload, share, and explore videos with a smooth, intuitive user experience
 - Real-time translator API, enabling support for 9 languages, enhancing accessibility for diverse audiences.
 - Gesture-based playback controls, and tiered viewing plans (Bronze, Silver, Gold)..
 - Technology Used: React.js, Node.js, Express.js, REST API, MongoDB, Redux
- **CookBook** | [LINK](#) February - April 2025
Recipe Management and Exploration Web Application
 - Built a responsive multi-page web application for exploring and managing recipes.
 - Collaborated in a team of 4 to design structured recipe pages with sections for ingredients, preparation steps, and nutritional information.
 - Developed the frontend using ReactJS, HTML, CSS, and JavaScript, ensuring clean UI and responsive design.

EXPERIENCE

- **Full Stack Web Developer Internship** April - June 2025
NullClass
 - * Built scalable full-stack web applications with responsive UI/UX.
 - * Managed real-time data with NoSQL databases and integrated Google OAuth authentication.
- **AI/ML Intern** January 2026 – Present
HePro
 - * Built and optimized machine learning models for real-time categorical prediction using preprocessing and feature engineering techniques.
 - * Worked in deadline-driven cycles, gaining hands-on experience in ML deployment and industry-oriented problem solving.

TECHNICAL SKILLS AND INTERESTS

Languages: C, Java, Python, JavaScript, HTML/CSS
Tools & Frameworks: ReactJS, Node.js, NiceGUI, Git, VS Code, MongoDB
Coursework: DSA, OOP.
Skills: Problem-solving, Collaborative Teamwork, Efficient project planning, Tech-adaptive.

POSITION OF RESPONSIBILITY

- **Team Leader – Deep Learning Project** SmartBridge Industry Project January– April 2025
 - * Led a 4-member team to design and implement an Early Disease Detection agent using deep learning.
 - * Organized milestones and deadlines, ensuring smooth collaboration and adherence to SmartInternz industry submission guidelines.
 - * Provided technical mentorship to teammates, resolving coding issues and guiding report preparation.
- **Avionics Engineer – CSE Domain** Team SAMMARD – VIT Vellore June 2025 – Present
 - * Contributed to software systems development for the student-led rocketry team comprising interdisciplinary engineering students.
 - * Collaborated on coding, simulations, and ground station software.